WeldCheck User Guide

A simple guide for recording a SMAW job, checking a weld photograph, understanding the result, and keeping a report.

What WeldCheck does

WeldCheck records mild-steel SMAW job details and checks the visible surface shown in one photograph. It helps you organize evidence. It does not certify a weld or show internal defects.

The full flow

1. **Create a job**
   Enter the sample number and the welding settings that were actually used.
2. **Complete the checklist**
   Mark each preparation and safety item as Done before welding.
3. **Add a weld photo**
   Take a clear photo after the weld has cooled and loose slag has been removed.
4. **Read and confirm the result**
   Review the visible observations, possible causes, actions, and limits.
5. **Keep the evidence**
   Open the report later, print it, or save it as a PDF.

Guest or Google?

Guest mode works immediately and keeps records private. Use Google if you want the same records on another device. Connecting Google keeps the guest records already created.

1. Create the welding job

Open **New job**. Use the values from the actual welding setup. The photograph cannot tell WeldCheck these values.

Required job details

Sample number	Your own short ID, for example WC-001.
Joint type	Butt, fillet, lap, or T-joint.
Plate thickness	Enter the measured thickness in millimetres.
Electrode	Enter the classification and size, for example E6013 and 3.2 mm.
Position	Flat, horizontal, vertical, or overhead.
Current	Enter the welding current in amperes.

Optional details

- Add a job name to make the record easier to find.
- Add the welder or student name when your project requires it.
- Use the note box for a short fact that belongs with the job.

Before you continue	Check the numbers once. They appear in the final report exactly as entered.

2. Complete the preparation checklist

Tap each item only after it is complete. WeldCheck saves progress while you work.

The nine checks cover

- Material and plate thickness
- Joint arrangement and clean surfaces
- Electrode classification and size
- Welding current
- Holder, cables, and ground clamp
- Helmet, gloves, and protective clothing
- A safe and properly arranged work area

Pending means stop

If an item is still Pending, complete it before continuing. The button will not move to the photo step until all nine items are Done.

3. Take a useful weld photo

1. **Wait and clean**
   Let the weld cool. Remove loose slag so the bead surface can be seen.

2. **Use even light**
   Avoid deep shadows, bright flash glare, and a very dark background.

3. **Fill the frame**
   Hold the camera about 15-20 cm away. Include the bead and both weld toes.

4. **Hold steady**
   Tap the weld on the screen to focus, then keep the phone still.

Photo quality comes first

WeldCheck checks brightness, contrast, sharpness, and image size on the device. A poor photo is stopped before the AI check and you are told how to retake it.

4. Understand the result

Read the result from top to bottom. Start with the status, then check the photograph and the evidence sections.

Acceptable appearance	None of the four supported visible conditions was returned. This is not a welding-code pass.
Visible issues	One or more supported surface conditions is visible in the photograph.
Retake photo	The photograph is not clear enough for a useful visible-surface check.

Supported visible conditions

- Visible porosity: rounded pits or holes visible on the surface.
- Undercut: a visible groove along a weld toe.
- Excessive spatter: many scattered metal droplets around the bead.
- Irregular bead: clearly uneven bead shape or width.

How to read the explanation

- Visible observations say what can be seen in the photo.
- Why this result explains how the observations support the status.
- Possible causes are suggestions only. A photo cannot confirm the cause.
- Recommended actions tell you what to check, correct, or photograph again.
- Limitations say what WeldCheck cannot decide.

Your review matters	Choose Result matches, Correct conditions, or Not sure. This keeps your own observation separate from the AI result.

5. Reports, history, and downloads

Open **Dashboard** to find past jobs. A recheck creates a new inspection and keeps the earlier result.

Print or save a report

- Open an inspection result.
- Choose Print or save PDF.
- In the print window, choose Save as PDF if you do not need paper.

Download a spreadsheet

Choose **Download spreadsheet** to create a CSV file. Open it in Excel, Numbers, or Google Sheets. It uses one row for each inspection.

Download a full backup

Choose **Download full backup** to create a JSON file. This file is for safekeeping or technical recovery. It is not designed to look like a report. Keep it private because it contains job and inspection details. Photographs are not placed inside the backup file.

Which download should I use?	Use the printable report for people. Use the spreadsheet for tables and project analysis. Use the full backup only to preserve the complete record data.

Need help?

Open the Help and guide page from the menu. It includes this guide, short answers, photo tips, and the meaning of every result section.

6. Account, privacy, and troubleshooting

Account and privacy

• Guest records are private and use a secure guest account.

• Connecting Google keeps the same guest records and allows access on another device.

• Each user can read only their own jobs, inspections, and photographs.

• The OpenAI and Supabase private server keys are never placed in the web page.

If something does not work

• Page still loading: check the internet connection and refresh once.

• Google sign-in does not return: use the same browser and allow the Google sign-in page to finish.

• Photo rejected: use more even light, clean the camera lens, and hold the phone steady.

• AI inspection unavailable: keep the job and try again when the internet service is available.

• Daily limit reached: wait until the next day or ask the project administrator.

Important limit	WeldCheck checks visible surface appearance in one photograph. It does not show penetration, internal defects, mechanical strength, or compliance with a welding code. Use suitable engineering tests and a qualified person when those decisions are required.

Quick reminder

Enter the real setup. Complete all checks. Take a clear photo. Read the evidence. Confirm what you observe. Keep the report.